Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Consider a discrete-time system having $A_d = 0.5, B_d = 0.1, \bar{x}_0 = 1,\ u_0 = 10, \Sigma_{\tilde{x},0} = 1$, and $\Sigma_{\tilde{w}} = 0.5$. What is $\bar{x}_1$?

1 / 1 point

1.5

Correct
Yes. This is correct.

2. Consider a discrete-time system having $A_d = 0.5, B_d = 0.1, \bar{x}_0 = 1,\ u_0 = 10, \Sigma_{\tilde{x},0} = 1$, and $\Sigma_{\tilde{w}} = 0.5$. What is $\Sigma_{\tilde{x},1}$?

1 / 1 point

0.75

Correct
Yes. This is correct.

3. Consider a discrete-time system having $A_d = 0.5, B_d = 0.1, \bar{x}_0 = 1,\ u_0 = 10, \Sigma_{\tilde{x},0} = 1$, and $\Sigma_{\tilde{w}} = 0.5$. What is the steady-state value of $\Sigma_{\tilde{x}}$? Enter your answer with two digits to the right of the decimal point.

1 / 1 point

0.6666667

Correct
Yes. This is correct.